

Example: Weight-Loss Program

10 participants show differences (after – before). After

removing 1 tie, $n = 9$ with 8 positives. Test H_0 : median diff = 0 at $\alpha = 0.05$ (right-tailed).

- $p = P(X \geq 8)$ with $X \sim \text{Bin}(9, 0.5) = P(X=8) + P(X=9) = 0.0195$.
- Since $p < 0.05$, reject H_0 ; median decrease > 0 .